

AI vs Real: A Deep Learning Approach to Identifying AI-Generated Images

Publisher: IEEE

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Abstract:

This research aims to explore the capabilities of several deep learning architectures in identifying AI-generated images, to verify the authenticity of digital media. Five pre-trained CNN architectures, namely VGG16, ResNet50, DenseNet121, EfficientNetB0 and MobileNetV2, were fine tuned to identify the best suited model to distinguish between real and generated images. Rigorous evaluation revealed that VGG16 was best suited for this task with 96 % accuracy. Grad-CAM was used to derive insights from the decision making process of the VGG16 model to help in understanding the model's predictions and enhance the confidence in the results by showing the regions of interest. These results highlight the potential of using deep learning models to detect synthetic visual media and thus support the development of effective authentication mechanisms against the rising levels of synthetic visual content.

Published in: 2025 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC)

Date of Conference: 16-18 May 2025

DOI: 10.1109/ASSIC64892.2025.11158087

Date Added to IEEE Xplore: 17 September 2025

Publisher: IEEE

▼ ISBN Information:

Conference Location: Bhubaneswar, India

SECTION I. Introduction

Rapid development in the generative adversarial network pushed the manner of synthetic image generation so far that this field does not know the difference at all between real and computer-generated images. Because of the adversarial training framework [1] applied to GANs, it is capable of producing hyper-real images indistinguishable from authentic photographs. While this has opened a lot of creative space in areas such as art, entertainment, and design, it also raised huge challenges in the safety and security of digital media, the spread of false information, and the authenticity of content. Its huge-scale presence of images generated by AI has caused much attention to how it may be misused in other areas: videos deepfaked, fake news distributed, and identities manipulated. These bring threats both to an individual and society as a whole.

Against this backdrop, the issue of authentic image authentication mechanism has emerged more conscious. Most of the traditional methods [2] of verification of images rely on simple pixel analysis. It has been found that modern AI generated images employ highly sophisticated techniques like detection techniques. Therefore, conventional methods are quite inapt for this. Deep learning models, especially convolutional neural networks (CNNs) have gained immense popularity as super tools for real or synthetic classification of images [18]. These models state to perform best on a number of diverse tasks like classification in images, object detection, and also image segmentation.

This work deals with investigating the performance of several existing CNN architectures that are pre-trained to identify and classify real as well as AI-generated images. For the purpose of this work, the CIFAKE dataset has been used which contains [3] an extremely large collection of real and synthetic images so that exhaustive experimentation and the evaluation of the models can be performed. In the research, five state-of-the-art CNN models are fine-tuned. These include VGG16, ResNet50, DenseNet121, EfficientNetBo, and MobileNetV2. All of them have unique architectural features like different depth, residual connection, and efficient designing principles that might affect whether or not they are able to detect subtle differences between authentic and AIgenerated content.

In sum, the main goal of this study is to examine which architecture is particularly well-adapted to distinguishing between real and synthetic images in terms of accuracy, robustness [4], and interpretability. The results of the experimentation add great value in understanding why each model is good at what it does

and therefore to improve the development of more reliable and efficient authentication systems [19]. Of these architectures, the best performance was observed for VGG16 with an impressive accuracy of 96 % on the evaluation data set.

This really demonstrates the promise of deep learningbased methods that would be able to fight this wave of deceiving visual content. Grad-CAM is used as the visualization technique to [5] help better understand what portions of an image contribute most to a model's predictions to further improve upon the interpretability of its decision-making process. Grad-CAM aids in developing an understanding of which parts of the VGG16 model it focuses on to differentiate the real and synthesized images, again stamping credibility to its predictions. Explainability methods, such as Grad-CAM, are very important in ensuring deep learning models don't become labeled as a "black box" but are instead viewed more like the tool for a rational decision-making process.

As such, the research presented here once again underlines the relevant importance of deep learning techniques in addressing the rising concerns posed by synthetic image generation. Second, from the results presented above, one can clearly observe how results obtained from CNN models can correctly classify real images from those AI-generated images while also stressing the relevance of explainability to improve model transparency. This work continues on the vision of the integrity of digital media by providing a comprehensive framework by which to develop trustworthy image authentication systems that may mitigate the risks from deceptive visual content.

This work is organized in [Section II](#) as it reviews the literature survey. [Section III](#) outlines the methodology, detailing its features and functionality. Results and discussion are found in [Section IV](#), where the effectiveness of the system is analyzed. Finally, [Section V](#) concludes with key findings along with future implications.

SECTION II. Literature Survey

Synthetic media, particularly AI-generated images, are extremely challenging as the task of distinguishing between real and fabricated content becomes very challenging. This led to further research on detection techniques, by focusing on digital watermarking, image registration, and deep learning approaches. Various techniques for unveiling fake images have been proposed recently using GAN-based models, convolutional neural networks, and spectral analysis for diffusion models. These advances are toward the development of media forensics as well as the inhibition of the diffusion of misinformation in the digital realm.

Digital watermarking began as an application for copyright protection but has [6] evolved into a vast number of applications, including the detection of fake news. The requirements for watermarking techniques detecting disinformation across multimedia content are analyzed through a meta-survey of the existing research and summarizing notable survey works.

Therefore, to address the differences in resolution and spectra causing some accuracy problems in the registration process, a new infrared and visible image registration algorithm is [7] proposed. The proposed VICycleGAN based on the algorithm is used for an image translation task, and then a normalization-based attention mechanism for effective feature extraction is included. This provides a promotion of about 5 % improvement in registration accuracy. The CIFAKE model examines AIgenerated synthetic images and attempts to identify them using a high-accuracy classification approach. This model employs the use of a convolutional neural network and gives explainable AI features, which reveal imperfections in the background are crucial in distinguishing real from AI-generated images [8].

Tackling the Growing Problem of Deepfakes: A Multi-model Framework, called MMGANGuard, is designed to reach [9] high-accuracy detection of GANgenerated images. ResNet50V2 and DenseNet201 are combined for more than 97 % accuracy in detecting deepfakes. New method for detection of synthetic images generated by diffusion models is proposed and demonstrated with spectral analysis again as a finding of the frequency artefacts in the residual [10] image. Successfully allowing for detection with mild jpeg compression, new techniques open avenues for media forensics.

To introduce a novel architecture for GAN, 3G-AN, for human visual perception modeling - that generates images in coarse, medium [11], and fine resolution stages, capturing the overall structure at coarse stages, and finetune finer details at medium and fine resolution stages. Their approach beats many state-of-the-art approaches on MNIST, CIFAR10, and Celebrity faces datasets, generating nearly anomaly-free very realistic images. Multimodal Fake News Detection Using Pooling and Attention Blocks Tackled the modality fusion

challenge by proposing a mechanism leveraging the spatial and temporal knowledge of improvement in detection with the baseline models that have evaluation on the Weibo dataset.

The proposed scheme integrates text and image features together for better reliability and interpretability to improve detection for fake news. Electra and XLnet pre-trained models are used [12] for text feature consideration, while EfficientNetBo is adopted for images. The results revealed better accuracy in detection along with enhanced interpretation of the model as compared to existing techniques. This work explores the state of the art about text-to-image synthesis, with a particular interest in deep learning models that should be able to generate realistic images given considerable textual descriptions. By inspecting the success of GANs and diffusion models. Some common datasets and performance metrics in this field are reviewed along with challenge statements and future directions.

This work proposes [13] a model for detecting fake news based on multi-modal features by integrating text and visual information in one system. It uses a framework that runs with a series of deep learning methods like BERT, GRU, and ResNet-CBAM followed by feature fusion and feature dimension reduction and records tremendous accuracy in datasets, like Weibo and Fakeddit. This work introduces a novel approach that takes advantage of residual noise, warping artifacts, and blur effects for detecting deepfake images. In terms of steganalysis, the model detects pixel-wise noise traces along with facial landmarks to capture unnatural deformations. The results show that the model is robust against various types of deepfakes.

The work presents an explicit mask-based unsupervised image-to-image translation technique controlling the image translations. The model changes part of the input images while retaining the general structure by manipulating a regularization term. This helps in ensuring translation flexibility and diversity. In this work, they apply conditional GANs to synthesize [14] progression images of pneumonia from chest X-rays, filling a gap between normal and pneumonia-infected images. This resulted in plausible progression images and validated the effectiveness of the proposed framework with the use of a conditional vector for domain change for its application in medical imaging.

A new approach that integrates features of multimodal is proposed to detect fake news [15] on social media platforms. Features extraction, fusion, dimension reduction, and classification are implemented with BERT, GRU, CNN, ResNet-CBAM, and FLN classifiers. The experimental results on some datasets such as Weibo and Fakeddit show significant improvements in the accuracy of detection compared with other methods. The best detection accuracy is as high as 98% for some datasets, showing the effectiveness of the proposed approach towards handling multi-modal data.

Proposed herein is a residual noise, warping artifacts, and blur effects generative method of detecting deepfake images. This method relies on residual-noise detection using the steganalysis network, landmarks in facial feature deformations, and measurements of image quality for blur effects. Experimental results described better performance of the network for detecting different types of deepfakes, making it tolerant to various manipulation methods. It thus stands to show that this proposed method is more stable and precise than existing deepfake detection networks.

This work presents [16] the construction of an artificial intelligence system, which is capable of detecting deepfake from more than one dataset. This study proposes neural network models based on deep learning using random forest (RF) and support vector machines (SVM) as classifiers for deepfake detection. The use of two classifiers (RF) and (SVM) and their combination with a convolutional neural network is the first study of its kind in the field of deepfake detection in images from three open-source datasets (FaceForensics++, FaceAntiSpoofing, and iFakeFaceDB). This proposed method shows an accuracy of 96%, 87% and 52 % in iFakeFaceDB, CelebA-Spoof, FaceForensics ++ and respectively.

This research proposes a method for extracting image content, classifying it, verifying its authenticity, and identifying manipulations. To verify the authenticity of the images, a method using a CNN deep learning model [17] was used. Pixel-based fake image detection found tampering of an image up to a certain level. The raw image was mangled in sections or as a whole, as it is necessary to recognize the type of image tampering and localize the tampered region. The image was converted into pixel format, and the RGB values were fed into the network's input layer. The output layer has two neurons for phony and genuine images, allowing us to infer if the image is phony and how likely it is to be tampered.

Existing synthetic media detection techniques often exhibit limitations including high computational complexity due to deep architectures, lack of model interpretability, and insufficient robustness across different types of synthetic image manipulations. The proposed system addresses these gaps by leveraging the established pre-trained CNN architectures, providing a balance between performance and computational cost suitable for practical applications. Critically, Grad-CAM is employed to generate visual explanations of the model's classification decisions, to provide crucial insights into feature extraction and decision-making processes, thus enhancing interpretability.

SECTION III.

Methodology

In this work, deep learning techniques are extensively used to identify real and AI-generated images in the presence of increasing proliferation of synthetic media. Utilizing the CIFAKE dataset, five pre-trained Convolutional Neural Network models-VGG16, ResNet50, DenseNet121, EfficientNetBo, and MobileNetV2 were respectively fine-tuned for good classification performance on the images, with the best performance seen from the VGG16 model. Additionally, Grad-CAM was employed for the interpretation of the model; this was to disclose some insight into the thought process behind the decision-making process. The methodology as in Fig. 1 explains the prospects that deep learning possesses in the fight against deceitful visual content.



Fig. 1:
Methodology flowchart

A. Data Acquisition and Preprocessing

The dataset used here is CIFAKE, which contains both real and fake images created artificially using AI. The CIFAKE dataset is a collection of low-resolution (32×32 pixels) images designed for AI-generated image identification tasks. It consists of two balanced classes: real images sourced from the CIFAR-10 dataset and synthetic images generated by a stable diffusion model. The dataset is balanced, ensuring an equal number of real and fake samples, making it ideal for fair classification. Its small image resolution emphasizes challenges in distinguishing real from AI-generated content. CIFAKE is widely used for benchmarking deep learning models in detecting synthetic images. The dataset was preprocessed by standardizing images of uniform dimensions and normalization of pixel values for inputting into models of neural networks. Multiple data augmentation techniques have been carried out to enhance diversity within the dataset and thus increase generalization for the models, including rotation, flipping, and scaling of images.

B. Model Implementation

This study used five pre-trained CNNs: VGG16, ResNet50, DenseNet121, EfficientNetBo, and MobileNetV2. These models have reached an excellent level of results on a wide range of computer vision benchmarks. They have stood the test of performance by showing the ability of their pre-training on the ImageNet dataset. These models were, with transfer learning, fine-tuned to adjust for CIFAKE type characteristics in classifying images as real or AI-generated.

C. Model Training

CIFAKE dataset images were classified as real or fake by fine-tuning pre-trained CNN models. Training used callbacks for early stopping, learning rate scheduling, and model checkpoints so that overfitting was well avoided and efficiency improved. VGG16, ResNet50, DenseNet121, EfficientNetBo, and MobileNetV2 were trained using random hyperparameters, but the base models performed better than those hyperparameter-tuned with no improvement in adjustments in major parameters.

D. Model Evaluation and Selection

The accuracy, precision, recall, and F1-score were used in the evaluation of the model's performance. The evaluation metrics were calculated on an independent test set that was not the same as that of training and validation. Following all these rigorous evaluations, it was noticed that VGG16 achieved the highest performance during the test. It correctly classified real images from the synthetic ones and had a proper classification accuracy of 96 %. This shows that the other models performed with a compromised type of classification, as VGG16 is much better for this type of classification.

E. Grad-CAM for Model Interpretability

This is applying the gradient-weighted class activation mapping method, which is known as Grad-CAM. It improves the transparency of the VGG16 model's decision-making process, giving visual explanations by highlighting certain regions in the image that influence the model's classification decisions. As such, one can understand how the model identifies distinguishing features between real and AI-generated images in those visualizations. In addition to supplying a more clear interpretation of the model's prediction, the use of GradCAM improves the credibility of the classification result.

Fig. 2:
Grad-CAM heat map

[Fig. 2](#) shows heat maps provided by Grad-CAM that revealed the model focuses on specific regions of the image, such as textures, edges, and background inconsistencies, to distinguish real from synthetic content, revealing patterns like artifacts or imperfections unique to AI-generated images.

SECTION IV.
Result and Discussion

In this experimental setting, various deep learning models such as VGG16, ResNet50, DenseNet121, EfficientNetBo, and MobileNetV2 were used for an image classification problem in CIFAKE dataset, and among these, the most efficient model was VGG16. The final model was trained on the CIFAKE dataset with 100000 images for training and 20000 images for testing. The preprocessing of image data was done using a rescaling factor of $1/255$. Additionally, a batch size of 32 was used in training. The architecture of the model also included Flatten, dropout for regularization, and sigmoid output layer for binary classification. Training was done with early stopping, along with reduction of learning rate on plateau and model checkpoint callbacks to optimize performance and prevent overfitting. The optimizer applied was Adam with a learning rate of $1e-4$, and the loss function applied was binary cross-entropy. The performance of the model on the test set was judged. The final model produced an incredible accuracy of 96 %, where it signifies its high sensitivity characteristic in classifying the real and fake images within the dataset, while losing 10 % as shown in [Table.1](#). It achieved a precision of 94 % for real and 99 % for fake images, showcasing its ability to accurately identify true positives in both categories. Also, it demonstrated a recall of 99 % for real and 93 % for fake images, displaying its sensitivity to detecting real images effectively but slightly missing some fake ones. The F1-score for both real and fake images was 96 %, emphasizing the balance between precision and recall in the model's performance.

Table. 1: Experimental results

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So far, the model appears to have had enough success in distinguishing real images from fakes in the CIFAKE dataset that it would be suggested as useful for similar classification tasks for images. Deep architecture existing in VGG16 played a major role in achieving the high results in the aspects of accuracy. This early stopping callback had prevented overfitting, and it reduced the learning rate on a plateau thereby making it possible to fine tune details at later stages of training to improve convergence. Model checkpoints have been helpful in preserving the best performing model during training through an efficient way not to lose optimum weight during the course of learning. An additional optimized Adam optimizer choice with a binary cross-entropy loss function also played key roles in the optimization processes, focusing upon the model while converging more of the information much more effectively and efficiently.

The batch size was set at 32 to balance up the training process that allowed the model to learn without overloading up the systems' memory or resources for the computation. The dropout layer introduces regularization, which ensures that there is no overfitting by enabling neurons to shut down randomly during training. This is especially helpful when working with large data sets like CIFAKE. The notable 96 % accuracy of the model shows that it applies quite nicely to real-world settings where the clear distinction between real and fake images is crucial. The model could be deployed in digital platforms to detect and flag AI-generated images, preventing their misuse in spreading misinformation or deepfakes. It can also be employed in content moderation systems to authenticate the originality of media shared online. However, the results indicate that in the final model there could potentially be a degradation of performance by 10 %, meaning there is room for improvement in dealing with challenging or adversarial examples.

SECTION V.
Conclusion

This research conducted an intense study in the paradigm of deepfakes, or images by AI. It has been shown that it is indeed possible to classify authentic versus fake images accurately using state-of-the-art convolutional neural networks (CNNs) such as VGG16, ResNet50, DenseNet121, EfficientNetBo, and

MobileNetV2, with VGG16 surpassing others with remarkably high accuracy of 96 %. The integration of Grad-CAM was truly insightful into the decision-making process of the model, hence increasing the interpretability and reliability of the results. The successful application of deep learning techniques in this study underscores the potential of artificial intelligence in combating the spread of misinformation and protecting the integrity of digital media.

Despite its success, the study also encountered challenges and limitations. The low resolution of the CIFAKE dataset posed difficulties for fine-grained feature extraction, which could be addressed by integrating higher-resolution datasets or multi-resolution analysis in future studies. Additionally, while the dataset is balanced, it may not encompass the full diversity of AI-generated techniques, limiting generalizability. Incorporating synthetic images from newer generative models can enhance the dataset's variability and significantly improve robustness. Furthermore, reliance on fixed hyperparameters may have constrained model performance, suggesting that more advanced optimization techniques could yield better results.

The future work could involve fine-tuning hyperparameters, experimenting with more developed augmentation techniques to expose the model to harder-to-classify examples, or possibly even the use of other architectures with improved feature extraction capabilities such as Transformer-based models to enhance robustness, precision and accuracy during classification. Crossvalidation can also be considered to evaluate the generalisability of the model, which would make sure the model performs consistently for any subset of the dataset. Additionally pre-trained models or ensemble methods can also be deployed to further reduce false positives, improving precision and accuracy, and enhancing the model's robustness against new unseen types of data.

Advanced optimization techniques such as mixedprecision training, pruning technique and knowledge distillation can also be explored to improve efficiency. Additionally, exploring hybrid approaches that combine VGG16 with other authentication techniques, such as digital watermarking or spectral analysis, could enhance robustness against adversarial attacks. Deployment in areas like social media, journalism, and forensic analysis can amplify its real-world impact, making it a cornerstone for addressing deepfake challenges. Overall, this study will be instrumental in providing a basis for making deep yet profound advances in the detection of deepfakes and, by inference, provide a foundation upon which further research and development shall continue in this important arena.

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